

Concurrent Aerobic and Strength Training for Performance in Soccer

27

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Soccer as a Concurrent Modality

Demands of Football

Elite soccer players cover 8–13 km during matches [1] with an average intensity near the anaerobic threshold (70–75% of VO_2max) [1–3], which suggests that performance at the elite level may, in part, be determined by aerobic fitness [2]. However, despite low- to medium intensity running being the predominant activity pattern of soccer players, high-impulsive efforts, such as sprints, jumps, duels, and kicking, require maximal neuromuscular efforts [4]. These have the goal of maximizing the impulse produced [5] as this determines the decisive decision-making situations in professional football (e.g., speed) [6]. Moreover, the acceleration and deceleration activity profile in soccer results in high neuromuscular strains, even when speed is low [7]. Nowadays, players are experiencing an increase in match-play physical demands in part due to shorter between-match recovery periods and high neuromuscular demands (e.g., greater number of high-intensity running actions and acceleration requirements) [2, 8, 9]. Competitive soccer places a considerable stress on the physiological and biomechanical load (e.g., stress in joints and muscles) adaptation pathways. It seems that match-play represents an important component of the strength/power training stimulus; indeed higher match exposure is associated with improved neuromuscular qualities such as jumping height and power [10–12].

In soccer, potentiated neuromuscular and endurance-related qualities provide a competitive advantage, as they are associated with improved fatigue resistance

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during the game [13, 14] faster post-match recovery [15] and injury prevention [16–18]. Therefore, players perform intense training programs (e.g., strength/power, endurance, speed endurance) aiming to improve neuromuscular and specific endurance performance.

Methodological Considerations

Recently, it has been highlighted that particular attention should be given to the selection of the modes of strength/power and endurance training [19]. It is believed that when they are at opposite ends of the biomechanical and neuro-coordinative spectrum, the anatomical and performance adaptations may be reduced, and the accuracy of the intended movement, fluidity, and elegance that characterize excellence may be compromised [19]. These methodological considerations call for the adoption of an integrated approach in training design (e.g., exercise, session plan), involving all technical staff such as the coach, the physical coach, and the physiologist. For example, two “physical conditioning” training exercises (e.g., $4 \times 4'$ speed running at 15 km/h vs. $4 \times 4'$ situational drill) may result in similar internal load (e.g., 90% maximal heart rate), but may differ in the highly impulsive requirements (e.g., more acceleration/decelerations during the drills). As so, manipulation of exercise structure is key for strength gains transfer to the high-impulsive actions required during soccer [19]. In fact, coaches adopt small-sided games to develop player’s endurance qualities [20], but usually, even more to prepare next game strategic plan and consolidate players and team technical-tactical behaviors. The later aspect comprises the greater proportion of training time in strategic sports. In fact, different game structures (pitch size and player numbers) may influence the frequency and fatigue development of high-impulsive actions [21, 22]. Consequently, the neuromuscular involvement during the “overall” training sessions is an important consideration in soccer. Notably, the physiological and molecular events associated with endurance fitness development and maintenance (expression of PGC-1 α mRNA) [23] are usually targeted during the typical soccer training (e.g., tactical exercise, technical circuits that often involve frequent displacements or small-sided game exercises performed during a 90-min soccer competition/training session) [19].

Research in Concurrent Training in Soccer

The physiological and performance adaptations to different CT program (Tables 27.1 and 27.2) have been investigated in youth (13–15 years), [24] young adults (17–19 years), [25, 26] and adult soccer players [27–30]. Generally, studies have examined the training effects of two sessions per week [24–27, 30]. Nevertheless, the effect of higher frequency of concurrent training sessions (i.e., three) [29] and a combination of different frequencies of endurance and strength session during the CT (endurance block composed of two endurance training sessions and one strength training session by week and *vice versa*) have been investigated [28].

Table 27.1 Physiological and functional adaptations to concurrent training on alternating sessions

Study	Level/Country/ <i>n</i> (age)	Type of training	D	P	Physiological adaptations	Performance changes
Nunez et al. [25]	Semi-professional/ Spain/16 (28 ± 3.7)	Strength and Endurance with 24 h rest between sessions; SE and END —a sequence of general, special, and specific exercises incorporated in different training blocks. ET block (2 sessions END + 1 session SET) SET block (1 session END + 2 session SE)	4 blocks of 12 weeks	S		↑73–80% Probst test ↑11.1–16.2% SJ ↑8–8.7% CMJ ↑6–7% CMJWAS
Wong et al. [24]	Professional/ Hong-Kong/9 (24.6 ± 1.5)	Strength in the morning and Endurance in the afternoon (5 h interval) SE : 5 exercises; high-pull, jump squat, bench press, back half-squat, and chin-up; 4 sets at 6RM with 3-min rest between sets) END : 16 × 15 s at 120% of MAS with 15 s rest	2×/ week/8 weeks	PS		↑4% VJ, ↑5.9% T _{10m} ↑2.8% T _{30m} ↑19.7% YYIR1, ↑3.1% MAS ↑9.2% MAS _{distance}
Makhlouf et al. [30]	Elite youth/ Tunisia/14 (13.7 ± 0.5)	Strength and Endurance with 24 h rest between sessions SE : traditional resistance exercises, olympic lifts, plyometrics, 3 sets, 5–10 reps) END : 2 × 12–16 runs at 110–120%MAS)	2×/ week/12 weeks	IS	↑40.1% 1RM _{Squat}	↑6.9% T _{10m} ↑5.7% T _{30m} ↑5.7% Agility _{15m} ↑13.6% SJ ↑8% CMJ ↑4.1% MAV ↑54.5% YYIR1 ↔5JT

(continued)

Table 27.1 (continued)

Study	Level/Country/ <i>n</i> (age)	Type of training	D	P	Physiological adaptations	Performance changes
Enright et al. [29]	Professional/ Premiere League Club/7 (17 ± 2)	Endurance and Strength and with 120-min rest between training components, SE: 5 traditional resistance exercises, 4 sets × 6reps 85% IRM) END: small-sided games, 4 V 4, 4 × 4 min/3-min rec, 85–95% maximal heart rate	2×/ week/5 weeks	IS	19.1%IRM _{Squat} * ↑27.3%IMVC LR ↑13.3%ConcQuads _{180°} ↑11–19.2%ConcHams _{60–180°} ↑23.1%EccHams _{120°} ↑14.3%AoP* ↔ MVIC _{peak force} , EccQuads _{120°} , ConcQuads _{60°} , FL, MT _{distal/} medial/proximal	↑8.2%SJ ↑5.6%10 m* ↔ T _{30m} , CMJ

* significantly different between groups [29]

Table 27.2 Physiological and functional adaptations to different within-sessions order concurrent training (strength plus endurance and endurance plus strength)

Study	Level/Country/ <i>n</i> (age)	Type of training	D	P	Physiological adaptations	Performance changes
Strength plus endurance						
McGaw/ley and Andersson [26]	Semi-and professional/ Sweden/9 (23 ± 4)	5-min rest between training components END: repeated sprint ability, speed endurance, high-intensity interval dribbling track and speed agility, and quickness circuits) SE: (olympic lifts and traditional resistance exercises, 2–3 sets/5–10 reps, 75–85% 1RM; plyometrics 3–4 sets/4–15 reps, core strength)	3×/ week/5 weeks	PS	↑6–7.6% fat (%. kg) ↑1.5–3% lean mass (%) and KG) ↑18.7% 1RM Half-squat ↑28.5% 1RM Lunge ↑97.3% Iliopsoas (°) ↑5.3% Hamstrings (°)	↑1.4% T ₁₀ ↑7% CMJ ↑1.1% Agility ↑1.9% RSA ↑19.6% perf dec RSA ↑15.4% YYIR2
Helgerud et al. [27]	Elite/Spain/23(25; range 20–31)	15-min rest between training components END: 4 × 4-min runs in a treadmill (5.5% inclination) 90–95% maximal heart rate separated by 3-min jogging at 50–60% maximal heart rate SE: 4 sets × 4 RM half-squats 90° with 3-min rest between sets	2×/ week/8-weeks	PS	↑8.6% VO ₂ max ↑3.7% RE _(11km,h) ↑52% 1RM _{Squat} ↑49% 1RM/BW	↑3.2% T ₁₀ ↑1.6% T ₂₀ ↑5.2% CMJ
Makhlouf et al. [30]	14 (13.7 ± 0.5)	15-min rest between training components SE: traditional resistance exercises, olympic lifts, plyometrics, 3 sets, 5–10 reps) END: 2 × 12–16 runs at 110–120% maximal aerobic speed	2×/ week/12 weeks	IS	↑47.2% 1RM _{Squat}	↑5.8% T ₁₀ ↑2.6% T ₃₀ ↑4.3% Agility _{15m} ↑78.6% YYIR1 ↑5.3% MAS ↔ SI, CMJ, 5JT

(continued)

Table 27.2 (continued)

Study	Level/Country/ <i>n</i> (age)	Type of training	D	P	Physiological adaptations	Performance changes
Endurance plus strength						
Makhlouf et al. [30]	Elite youth/ Tunisia/15 (13.7 ± 0.5)	15-min rest between training components SE: traditional resistance exercises, olympic lifts, plyometrics, 3 sets, 5–10 reps) END: 2 × 12–16 runs at 110–120% maximal aerobic speed	2×/ week/12 weeks	IS	↑40.5% 1RM _{Squat}	↑3.2–4.3% T ₁₀ and T ₃₀ ↑1.6% Agility _{15m} ↑7–11% SJ and CMJ ↑4.7% MAS ↑58.8% YYIR1 ↔5JT
McGawley and Andersson [26]	Semi-and professional / Sweden/9 (23 ± 4)	5-min rest between training components END: repeated sprint ability, speed endurance, high-intensity interval dribbling track and speed agility, and quickness circuits) SE: (olympic lifts and traditional resistance exercises, 2–3 sets/ 5–10 reps, 75–85% 1RM; plyometrics 3–4 sets/4–15 reps, core strength)	3×/ week/5 weeks	PS	↑5.2–7.1% fat (% and kg) ↑1.6–3.6% lean (% and KG) ↑19.1% 1RM Half-squat ↑19.1% 1RM Lunge ↑165.2% Iliopsoas (°) ↑10.3% Hamstrings (°)	↑2.2% T ₁₀ ↑1.9%CMJ ↑0.9% Agility ↑0.8% RSA ↑16.8% perf dec RSA ↑22.9% YYIR2
Lopez- Segovia et al. [28]	Elite/Spain/19 (18.4 ± 0.6)	SE + END SE: jumps with and without external loads, traditional resistance exercises. Speed agility and quickness exercises with and without external loads END: high-intensity runs, physical- technical circuits and small-sided games, with maximal intensity during 4–6-min periods.	2×/ week/16-weeks	PS- IS	↑5.8%Fsquats _{20kg} ↑7.1% Fsquats _{30kg} ↑5.2% Fsquats _{40kg}	↑6.8%CMJ _(20kg) ↓ 2.3%T ₂₀ and T ₃₀ ↓3.2%T _{s/10–20-} ↓1.6%T _{s/10–30-} ↓2.6%T _{s/20–30-}

Enright et al. [29]	Professional/ Premiere League Club/8 (17 ± 2)	SE + END with 30–45-min rest between training components SE: 5 traditional resistance exercises, 4 sets × 6reps 85% 1RM) END: small-sided games, 4 V 4, 4 × 4 min/3-min rec, 85–95% maximal heart rate	2×/week/ 5 weeks	IS	↑10.3% 1RM _{Squat} [*] ↑20.3%1MVC _{LR} ↑2.6%ConcQuads _{180°} ↑9–12%ConcHams _{60–180°} ↑16.5%EccHams _{120°} ↑7.5%AoP [*] ↔MVIC _{peak force} [*] EccQuads _{120°} , ConcQuads _{60°} , FL, and MT _{distal, medial and proximal}	↑7.5%SJ ↔ T _{10m} [*] , T _{30m} [*] CMJ
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↑ significant improvement, ↓ significant decrement, ↔ no significant alterations, ~ approximately, NS not specified, F/D frequency and duration of training protocols, P period of the soccer season, SE strength training, END endurance training, SJ squat jump, CMJ countermovement jump, CMJ/WAS countermovement jump with arm swing, MAS maximal aerobic speed, VJ vertical jump, Fsquats_(20–40kg) speed of movement during full squats exercise (range of the external load); T_{5–30m} sprint performance; T_{30/10–30} sprint performance in predetermined split distances; VO₂max, maximal oxygen consumption; RE_(11 km h) running economy (velocity); 1RM_{HS}, one repetition maxim in half-squat strength exercise; 1RM/BW, strength per kg of body weight; Rec recovery, C/S continuous jumps with legs extended, YYIRJ Yo-Yo intermittent recovery level one, MAS_{distance} maximal aerobic distance, SSG small-sided game, CMJ_(20kg) countermovement jump (external load), SAQ speed, agility, and quickness, HRmax maximal heart rate, P5 performed during preseason, IS performed during in-season, RSA repeated sprint ability, PT plyometric training, perf dec RSA performance decrement in the repeated sprint ability test, YYIR2 Yo-Yo intermittent recovery level two, FL Fascicle length, MT muscle thickness, AoP Pennation Angle; Traditional resistance exercises (e.g., multi-joint closed-chain exercises such as squats, dead lifts, lunges)

^{*}significantly different between groups [29]

The training-induced adaptations of CT performed in alternating sessions (e.g., strength training on Tuesday followed by endurance on Wednesday or endurance in the morning and strength in the afternoon, Table 27.1) [24, 26–28], or within the training session (strength plus endurance-based drills or endurance plus strength-based drills, SE + END, and END + SE, respectively) [24–26, 29] have been investigated (Table 27.2 and Fig. 27.2).

Strength/power training component of CT involved isolated (e.g., closed-chain multi-joint traditional resistance exercises, TRE) or a combination of distinct movement patterns (e.g., TRE, ballistic exercises, plyometrics, and sport-specific force-based actions within the same workout) with distinct loads (e.g., 4RM, 6RM, and 75% 1RM) and a wide range of movement velocities to improve player's performance in relevant motor tasks (Tables 27.1 and 27.2). Endurance-based element consisted of isolated (high-intensity aerobic training, HIA) or a combination of high-intensity training methods (e.g., HIA + speed endurance + repeated sprint ability, Tables 27.1 and 27.2). These loading schemes involved different intensity ranges (e.g., % of maximal heart rate and % of maximal sprinting speed) and exercises formats/structures (running based, dribbling tracks and small-sided games). Moreover, the independent variables (e.g., intensity, frequency, volume, work:rest ratio) of the strength or endurance loading scheme were manipulated in different ways, resulting in different training stimulus (Part II). Being so, players were exposed to a wide range of CT combinations (Tables 27.1 and 27.2). Small differences in the manipulation of the different mechano-biological descriptors of strength/power stimulus (e.g., load magnitude, volume) may result in distinct performance adaptations in soccer players [19]; determines the responses of cellular and molecular signaling pathways [31] (Part II). Consequently, the drawing of precise conclusions regarding the role of the different individual CT variables results in a complex equation. This is related to: (1) the wide variety of adopted training methods; (2) the short-term duration of the interventions; (3) the distinct season time lines used throughout the pre-season [25, 27–30] and/or in-season periods [24–26, 28]; (4) the different weekly loading schemes applied (endurance vs. strength); and (5) the different players' standard.

In fact, from the authors' knowledge just one study detailed the overall exposure (e.g., total distance covered, volume) during the exercise intervention [26]. Nevertheless, as discussed below, independently of the manipulation of the different concurrent variables, CT programs may result in substantial short-term physiological and performance improvements (Tables 27.1 and 27.2). Moreover, the magnitude of physiological and performance enhancements is comparable to the ones reported after strength/power studies conducted in soccer [19].

Physiological and Performance Adaptations to Concurrent Training in Soccer

Data related to the players' physiological parameters (e.g., % 1RM) and performance parameters (e.g., soccer-specific endurance tests and jump tests) were extracted and presented as the percentage of change, training efficiency, and corrected effect sizes (see Appendix for formulas).

Physiological Adaptations

The effectiveness of a training program is evaluated by the magnitude of sport-specific improvements (e.g., jumping, sprinting, and intermittent running capacity). Our analysis suggests that CT may result in improvements in different motor tasks and performance qualities in high- and low-level players (Tables 27.1 and 27.2 and Fig. 27.1). CT promote improvements in body composition (increased lean mass and decreased body fat) [29] endurance-related (VO_2max and running economy) [30] and neuromuscular-related characteristics (e.g., 1RM, rate force development) [24–26, 29, 30]. In fact, independent of the players' standard, enhanced (large magnitude) dynamic maximal force production [24, 26, 29, 30] can be obtained during different concurrent approaches [24, 26–30] (e.g., alternative days and within-session order). Specifically, increases in 1RM were observed during the performance of bilateral (Squat, Fig. 27.1) [24, 26, 29, 30] and unilateral (lunge) multi-joint exercises [29]. CT may also result in increases in relative maximal strength and peak torque during knee extension and flexion tasks both during concentric and eccentric isokinetic muscle actions [26]. Particularly, the increase in eccentric hamstring strength capabilities is especially relevant given the high rate of injury and subsequent recurrence that target this specific “vulnerable” muscle in soccer [32–34].

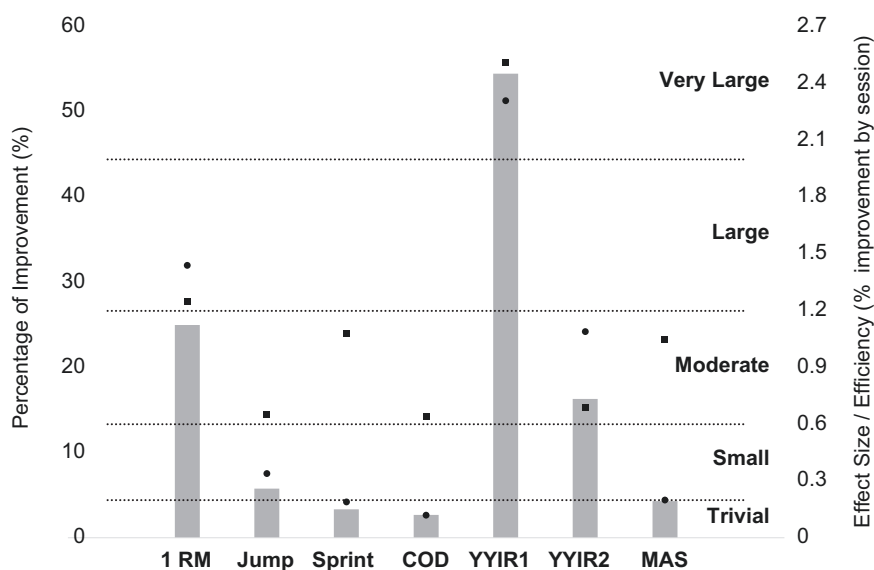


Fig. 27.1 Performances gains (solid bares), training efficiency (% improvement by session, circles), and magnitude of change (Weighted Effect Size, ES, squares) in 1 RM [24, 26, 29, 30], Jump [24, 26, 27, 29, 30], Sprint [24, 26, 27, 29, 30], COD [24, 29], YYIR1 [24, 27], YYIR2 [29], and MAS [24, 27] after short-term concurrent training interventions (5–12 weeks); studies included had to report changes in 1 RM. Training efficiency = % performance improvement/number of training sessions; 1RM one repetition maximum, COD change of direction ability, YYIR1 Yo-Yo intermittent recovery test level 1, YYIR2 Yo-Yo intermittent recovery test level 2, MAS maximal aerobic speed. Dashed horizontal lines represent the threshold values for corrected effect sizes (see Appendix)

Performance Adaptations

Pooled data from different studies (Fig. 27.1) [24–30] reveals an enhancement of neuromuscular performance (moderate magnitude) when measured by the performance of different multi-joint tasks such as jumping (SJ, CMJ, and CMJ-WAS), sprinting (10–30 m) [24, 26–30], and COD abilities (*t*-test and agility15 s test) [24, 29]. Although a similar magnitude of change is observed between these different motor tasks, the training efficiency is greater for jumping abilities followed by sprinting and COD abilities (Fig. 27.1). There are systematic [19] and meta-analytic evidences [35] showing that strength increases during the performance of traditional resistance exercises (e.g., squats and lunges) transfer positively to the performance of sprint-and-jump-related actions [19, 35]. In fact, associations between the measures of maximal (1RM) [36] and relative strength (1RM/BM) [37], as well as between mechanical properties of specific muscles (e.g., quadriceps), such as peak torque [19, 38] and peak power [39], and the ability of soccer players to perform complex multi-joint dynamic movements (e.g., jumping and sprinting actions) have been reported [11, 36–39]. Additionally, CT results in substantial gains in the performance of nonspecific (Fig. 27.1, Probs-test and MAS, extremely large and moderate magnitudes, respectively) [24, 27] and specific endurance tests (Fig. 27.1, YYIR1 and YYIR2, very large and moderate magnitudes, respectively) [24, 27, 29]. Notably, these performance improvements are relevant to optimize soccer players' match performance. Improved neuromuscular capacity (e.g., strength and power) and soccer-specific endurance (assessed by the Yo-Yo tests) have been linked to a greater fatigue resistance (e.g., during a worst-case match scenario) [13, 14], faster post-match recovery (e.g., lower match-related muscle-damage markers) [15, 40] and injury prevention (e.g., offers greater injury protection when players are exposed to rapid changes in workload) [16–18].

Considerations for Concurrent Training Design

There is a multitude of potential CT variables (within-session exercise order, between-mode recovery length, endurance training intensity, and endurance training volume) that may play a role in the acute and chronic interference [41] consequently influencing the chronic adaptations (Part II).

Within-Session Order

Clubs' obligations may involve team travelling and competing frequently within the in-season, limiting structured training and recovery opportunities within this important period [42]. The limited time for training results in coaches prescribing training sessions involving the training of different physical components (strength and football-specific endurance training) within the same day (e.g., morning and afternoon sessions during training camps) or within the same session. Nevertheless,

it's believed that the order in which these exercise modes are performed may potentially modulate interference and so, result in different training stimuli [41]. Although scarce, researchers have examined the short-term influence of SE + END [24–26, 29] and END + SE [24, 29, 30] within the same session. Generally, both forms of within-session order result in substantial improvements in neuromuscular performance assessed by means of 1RM (squat lunge and knee extension and flexion tasks), jumping, sprinting, and COD abilities (Table 27.2, Fig. 27.2b). Particularly, pooled data of the studies measuring strength gains revealed that END + SE results in a greater magnitude (Fig. 27.2a) of change in 1RM, acceleration (10 m sprint), and COD actions. Nevertheless, both organizations resulted in similar magnitude of gains on CMJ performance. The superior training effect of END + SE in sprinting abilities is further confirmed when pooling the different sprint distances (Fig. 27.2b). Notably, sprint “training efficiency” scores, representing the percentage of improvement by session, is higher for the END + SE

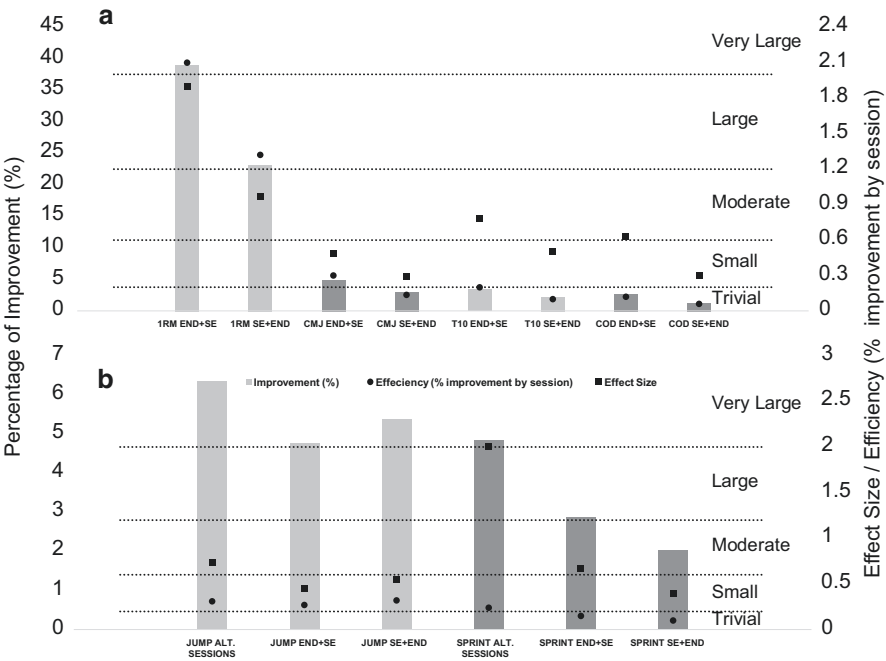


Fig. 27.2 Performances gains (solid bars), training efficiency (% improvement by session, circles), and magnitude of change (Weighted Effect Size, ES, squares) after (a) Concurrent training within-session order END + SE [24, 29, 30] and SE + END [24, 26, 29] within the same session (5–12 weeks). (b) Concurrent training performed in alternating sessions [24, 26–28] (5–12 weeks) and within-session order (5–12 weeks). Training efficiency = % performance improvement/number of training sessions, END + SE endurance plus strength, SE + END strength plus endurance, 1RM one repetition maximum, CMJ countermovement jump, T10 10 m sprint time. Dashed horizontal lines represent the threshold values for corrected effect sizes (see Appendix); Note: data extracted from study [28] are from the second season block (SE—6 sets/8 RM/fast loads, END—6 medium extensive intervals of 8 min)

independently of the method of analysis (10 m sprint vs. sprinting pooled data, Fig. 27.2a, b).

Almost 90% of all muscle injuries of professional soccer players are localized in lower limbs [34] with hamstrings muscle strains accounting for 12–16% of all injuries [34, 43, 44]. Particularly, the injury rate is higher in the last parts of the soccer matches [45]. Eccentric hamstring strength is a fundamental etiological factor associated with hamstring muscle strains [46]. As so, researchers investigate the short-term [46, 47] and acute [48] timing-induced effect (PRE vs. POST training session) of eccentric-based strength exercises (Nordic hamstrings exercises, NHE) targeting this soccer-specific “vulnerable” muscles. Lovell and colleagues [47] examined the short-term (2 times a week during 12 week) timing-induced effect of NHE within the session order. The authors observed that training adaptations were independent of the within-session exercise order; similar gains in strength, flexibility, architectural (penetration angle), and morphological changes (muscle thickness) were observed. Nevertheless, Small and colleagues [46] applying an identical research design in a shorter exposure period (8 weeks, 2 times a week, 3 sets of 12–10–8 reps) observed that the performance of NHE exercises at POST significantly reduced the negative influence of fatigue during soccer-specific exercise; players significantly increase eccentric hamstrings peak torque and the functional ratio (eccentric hamstrings; concentric quadriceps) at half-time (45 min) and post (90 min) soccer-specific simulation protocol (SAFT90) compared to the PRE group. However, PRE had a greater improvement in eccentric hamstrings strength and functional ratio (eccHams:concQuads) after the training intervention (assessments performed prior the soccer-specific protocol). Lovell and colleagues [48] investigated the acute neuromuscular and performance responses to NHE completed PRE vs. POST training session. The authors observed that PRE increased sprint performance but had a greater fatiguing effect on hamstrings strength (decreased eccentric peak torque) during 60-min soccer-specific exercise (peak torque assessed in 15-min intervals) [48].

Therefore, this greater fatiguing effect may render the players more susceptible to hamstring strain injury acutely during the remaining training session [48]. Specially, a higher injury predisposition might occur in the high-loaded players. Accordingly, the external load completed by players is highly variable within and between soccer-specific exercises, such as small-sided games and team strategic combinations. Different tactical and technical roles dispose to a high inter-player variability in specific locomotor activities [49, 50]. These previous factors (fatigue responses and inter-player variability within a session) along with the similar or greater gains of POST organization should be considered by coaches when defining the timing of the strength component. Nevertheless, it is important to highlight that these previous studies [46–48] were performed with amateur soccer players that are subject to a smaller training exposure (2–3 sessions a week) than professional players [46–48]. We believe that soccer-related technical staff should adopt an integrated approach when defining the exercise timing of the strength-based element of the session. Some of these are:

- Is the player returning from injury or not?
- Is the target increasing player’s maximal strength or strength endurance?

- Is the team in a congestive schedule period or not?
- What is the expected metabolic/mechanical stress of the “overall” session?
- Does the player show enough technical competency to perform a complex strength exercise in fatigued state?

Between-Mode Recovery Length (Proximity)

The recovery length allowed between concurrent exercise sessions is another important practical consideration [41]. Increasing training time available, as the observed within specific periods of the season (e.g., in-season periods with one competition per week or during pre-season and transition period) may allow coaches to implement isolated mechanical-and metabolic-based sessions [42] on alternating days [24, 28] or within the same day (morning and afternoon training sessions) [26, 27]. It's believed that an increase in recovery length will allow to not compromise the resistance training stimulus (via residual fatigue and/or substrate depletion) and anabolic response (via molecular interference, Part II) that is thought to occur when undertaking concurrent exercise sessions [41]. Enright and colleagues [26] observed small but substantial improvements (larger corrected effect sizes, ESs) in physical performance parameters in the END + SE vs. SE + END concurrent organization. Importantly, the recovery time allocated between the training of the two physical components was very different between exercises modes. The time allocated between the END + SE was of 120 min and of 45 min between the SE + END organization. As so, the authors attributed the superior results of the END + SE organization to the higher recovery time allocated between training bouts and the superior nutritional arrangements that could be applied when a greater interval period occurs [26]. The relevance of the between-mode recovery length is further confirmed in the study of Wong and colleagues [27] (5 h recovery between the strength and endurance session). In this study [27], were observed the greater improvements in jumping (large ESs) and sprinting abilities (very large ESs) among the seven selected studies. This is particularly relevant given that in professional football unexpected changes in team's training schedule occur that alter the organization of CT and the timing of food intake around each exercise session [51]. Nutrition plays a key role in modulating the physiological adaptations to CT (Part III) [51].

CT performed in alternating sessions [24, 27, 28, 52] may result in greater magnitude of changes in overall sprint and jump performance (Fig. 27.2b) than within same session arrangements [24, 29, 30, 51] or within the day [26]. This greater magnitude of change could be explained not only by a decrease in the interference effect (e.g., decrease training proximity) but as well as by a possible superior organization of the overall training session. Coaches may be able to structure the individual training sessions with the strategic and technical-tactical exercises that are dependent of strength or endurance qualities. This type of organization (strength/power vs. endurance-based sessions) may allow a greater input in the independent physiological and biomechanical load pathways of neuromuscular or endurance-related adaptations as well as in other relevant factors of soccer performance (technical-tactical) [42].

Intensity and Volume

Another practical consideration is the intensity and volume of the endurance component applied in the CT [53]. It has been suggested that given the conditional concurrent nature of soccer, concurrent high-intensity strength (high load few repetitions), and high-intensity endurance training modes (e.g., speed endurance) may enhance a player's overall performance [19]. In this regard, CT studies have mostly analyzed different combinations of these training methods (Tables 27.1 and 27.2).

A combination of resistance exercises, plyometric, and sport-specific strength exercises (e.g., accelerations and deceleration drills) is recommended to target a broad range of the force–velocity spectrum [19, 42]. This type of approach has been suggested not only due to its superior efficiency but also due to its ecological value [19]. Coaches should be conscious that a reduced transfer of strength gains to high-impulsive actions of soccer may occur when the prescribed exercises have a “poor” biomechanical specificity [54]. In this regard, open-chain, isolated and machine-based exercises (i.e., leg extension, leg press,) may target differently the intra-muscular and inter-muscular aspects of athletic performance; they may not provide adequate movement pattern specificity for optimal performance improvements in closed-chain sporting movements (e.g., sprinting and COD) [55]. Given the scarcity of studies in CT in soccer, an analysis of strength training studies performed in soccer may allow to draw better conclusions regarding the efficiency of different strength and endurance training methods. In fact, the majority of strength training studies in soccer have been performed during the pre-season period (for references, see [19]. The detraining during the transition period results in large reductions in endurance-markers (e.g., VO_2max and time to exhaustion) and performance of soccer-specific endurance exercise (Yo-Yo intermittent endurance exercise level 2) [42]. Therefore, a substantial amount of endurance-based work is prescribed during pre-season [42]. In this regard, Bogdanis and colleagues [56, 57] analyzed the effects of high-repetition/moderate-load (hypertrophy) and low-repetition/high-load (neural adaptations) programs on anthropometric, neuromuscular, and endurance performance; a considerable amount of interval training and small-sided games were performed [56, 57]. The hypertrophic mode was associated with increases in lower limb muscle mass, while the neural mode was more effective in improving relative strength, sprint, and COD performance, running economy, and fatigue resistance measured with a repeated cycle ergometer sprint test [56, 57]. Other researchers [58] found that explosive-strength training performed in parallel with endurance training resulted in improvements in the individual anaerobic threshold and neuromuscular parameters (e.g., low-force portion of the load-vertical jumping curve). These results suggest, at least in part, a better efficacy of neural-based programs in high-level players [56, 57, 59].

High-intensity endurance training modes (single and combined forms of HIT, Tables 27.1 and 27.2) have been suggested to better fit in the soccer training periodization puzzle than the adoption of high volumes/low intensity endurance modes [19]. This rationale is based on the following observations: (1) the acute anabolic

(e.g., growth hormone and insulin-like growth factor-I) [60–64] and redox environment; (2) HIT maintains a muscle fiber phenotype associated with strength and power capabilities [65]; (3) commonality of muscle action (brief and intense muscle contractions) [19]; (4) the adaptations in skeletal muscle and improvements in laboratory and field endurance-related parameters that are comparable to the observed after high-volume endurance training [66–71]; (5) increase in endurance and neuromuscular-related outcomes [72]; and (6) may improve muscle power-based actions [71–73]. Nevertheless, independent of the HIT training mode, exercise design should recreate the high-impulsive structure/nature of soccer (e.g., acceleration and deceleration, multidirectional running patterns).

Summary

Research in concurrent training in soccer reveals similar physiological and performance adaptations than the typically observed after strength/power training studies in soccer. Concurrent high-intensity strength and high-intensity endurance training modes may represent the most efficient combination to implement in soccer periodized training program. Within a CT, the observed magnitude of change and training efficiency seems as follows: alternating sessions>endurance plus strength within the same session>strength plus endurance within the same session. As discussed in this chapter, coaches should adopt an integrated approach considering the exercise timing of the strength element of the session (e.g., individual player, exercise structure, training methods, performance constructs, moment of the season). In fact, soccer is a concurrent sport by nature, due to the associated-endurance related demands triggered by the training of the other relevant constructs of performance (technical-tactical) and competition format (volume and intensity).

Appendix: Analysis and Interpretation of Results

To evaluate the magnitude of the effects, percent change was calculated for each dependent variable for each study using the below equation:

$$\left[M_{\text{post}} - M_{\text{pre}} \right] / M_{\text{pre}} \times 100 \quad (\text{Eq. 27.1})$$

where M_{post} was the post-training mean and M_{pre} the baseline mean. ESs (Effect size) were computed to present standardized training-related effect on the outcome variables [74]. The different ES within individual studies were calculated with Cohen's d , by dividing the raw ES (difference in means) by the pooled standard deviations, as proposed by Bornstein et al. [75] as followed (Eq. 27.1):

$$\text{ES} = g = \frac{(M_{\text{post}} - M_{\text{pre}})}{\text{SD}_{\text{pooled}}} \quad (\text{Eq. 27.2})$$

SD_{pooled} is the pooled SD of the measurements and was calculated as follows (Eq. 27.2):

$$SD_{\text{pooled}} = \sqrt{\frac{((n-1) \times SD_{\text{Pre}}^2 + (n-1) \times SD_{\text{Post}}^2)}{(2n-2)}} \quad (\text{Eq. 27.3})$$

where SD_{Pre}^2 is the standard deviation of the performance test completed before the training intervention and SD_{Post}^2 is the standard deviation of the performance test completed after the training intervention. To account for possible overestimation of the true population, ESs were corrected accounting for the magnitude of the sample size of each study [76]. Therefore, a correction factor (CF) was calculated as proposed by Hedges and Olkin [76].

$$CF = 1 - \frac{3}{4df - 1} \quad (\text{Eq. 27.4})$$

where $df = n - 1$. The corrected ES was calculated as follows:

$$\text{Corrected } ES_c = g \times CF \quad (\text{Eq. 27.5})$$

Threshold values for ES_c were defined as trivial (<0.2), small ($0.2-0.6$), moderate ($0.6-1.2$), large ($1.2-2.0$), and very large (>2.0) [77].

Training efficiency for each dependent variable was calculated as follows:

$$\text{Efficiency} = \frac{\% \text{ of improvement } \left[\frac{M_{\text{post}} - M_{\text{pre}}}{M_{\text{pre}}} \times 100 \right]}{\text{Number of training sessions (number of weeks} \times \text{number of sessions per week)}} \quad (\text{Eq. 27.6})$$

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